

Threat defence

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The year 2007 was characterised by unprecedented innovation and adaptability in the realm of security threats, as criminals once again demonstrated their ability to continually evolve attack strategies to keep ahead of even the most advanced human and technical defences.

Rogue developers create such threats by using worms, viruses, or application-embedded attacks. Botnets can be used to seed an attack, for example, rogue developers can use worms or application-embedded attacks, that is an attack that is hidden within application traffic such as web traffic or peer-to-peer shared files, to deposit "Trojans". This combination of attack techniques — a virus or worm used to deposit a Trojan, for example—is relatively new and is known as a blended attack. A blended attack can also occur in phases — an initial attack of a virus with a Trojan that might open up an unsecured port on a computer, disable an access control list (ACL), or disarm antivirus software, with the goal of a more devastating attack to follow later.

The evolving perimeter

Changes in network architectures and evolving threats create new security challenges. Also the concept of the network perimeter is changing. In the past, users could only access the network through a few ingress or egress points — usually where the internet connected to the enterprise network. Enterprises stacked security at the internet perimeter using firewalls and intrusion detection systems. However today, many more means of gaining entry to the network exist. With the perimeter having been extended and distributed, security too needs to be applied at each of these new ingress and egress points to avoid damaging threats, thus complicating security architectures.

Take for example, virtual private networks (VPNs). These allow enterprise users remote access to the corporate network and are much more widely used than just a few years ago. While previously enterprises might have insisted that VPN software run on a specific enterprise-configured computer, today users run VPNs from their own PCs or even from kiosks at copy centres or other businesses. This phenomenon allows many more entryways to the enterprise network and presents a significant challenge to IT departments. Is the computer equipped with virus protection? Are

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visitors / employees bringing their non-corporate assets into corporate network?

Or for that matter, let's look at wireless. Wireless LANs (WLANs) pose additional security challenges. Users operating on an unsecured wireless network at a local coffee house may be unaware that a rogue PC, also using the same wireless subnet, is depositing a virus on the PC. When that PC is later docked into the corporate network, the virus/Trojan could gain entry to the network.

Security — the reverse Moore's Law

As the network is becoming more vulnerable to attack because of the evolving perimeter, the threats themselves are changing. In addition to Trojans and botnets, newer, even more dangerous threats lurk. Two of the most troublesome are flash threats and self-mutating worms. Flash threats are so named because of the speed with which viruses or worms can spread. In 1999, a virus dubbed "Melissa," one of the earliest and most widespread viruses at the time, took 16 hours to spread globally, according to Network Associates Inc. In January 2003, the "Slammer" virus managed to infect more than 90 per cent of the vulnerable hosts worldwide within 10 minutes using a well-known vulnerability in Microsoft's SQL Server. New viruses in the coming months and years are expected to spread even faster. Therefore whatever defences organisations create, they must be able to identify the threat and respond much more quickly than ever before.

The other looming threat is the self-mutating worm. Today's worms are relatively unintelligent. They are programmed to follow a specific set of instructions, such as to infiltrate one machine through a specific port and once on the machine compromise it in some way, for example, causing a buffer overflow and planting a Trojan. If anything interferes with these planned instructions, the worm lacks the ability to adjust and dies. Now, however, rogue developers are adding intelligence and logic to worms so that if they can't complete a specific task worms can mutate and pursue other lines of attack.

Experts call this security dilemma "Moore's Law in reverse". Whereas Moore's Law postulates that processor performance will double every 18 months while costs will decline dramatically, security is moving in the opposite direction—networks are becoming less secure while the cost to defend them is increasing. This prognosis is supported by mi2g, a research firm in the UK that specialises in computer security. Mi2g reports that the economic dam-